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Zusammenfassung

Ein Saint-Venantsches Prinzip für eine gewisse Art von Randwertproblemen elliptischer Differentialgleichungen zweiter Ordnung wird bewiesen. Der Beweis erfordert die Abschätzung verschiedener energie-artiger Funktionale, die zur Herleitung von lokalen Schranken für die Lösung erforderlich sind. Zur Veranschaulichung wird das Prinzip auf die Berechnung der Kanalströmung einer zähigkeitslosen Flüssigkeit angewendet.

Inverse Deformation Results in Finite Elasticity

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1. Introduction

The equations of equilibrium for a finite deformation of an elastic body can be written in alternative forms with either the initial particle coordinates or the final coordinates as independent variables. For an elastically homogeneous material and zero body force, we show in Section 3 by a simple transformation that the two sets of equilibrium equations can be written in forms which are identical if initial and final coordinates are interchanged and if the strain energy W_0 per unit volume of the undeformed body is interchanged with the strain energy W per unit volume of the deformed body. If the material is incompressible, the pressure function p in one set of equations is replaced by $-p + W_0$ in the other set. This dual property of the elastic equations implies that given an equilibrium deformation with zero body force for one material, the inverse deformation (with initial and final coordinates interchanged in the description of the original deformation) is an equilibrium state for a second material, with strain energy related to the strain energy of the first material. The duality has been established previously by ADKINS [1][1]) for the particular case of an incompressible, transversely isotropic material in finite plane strain superposed on uniform extension, and Adkins deduced inverse solutions from equilibrium solutions for the same material.

[1]) Numbers in brackets refer to References, page 499.

If the inverse of every equilibrium state with no body force for the first material is an equilibrium state with no body force for the second material, then both materials are isotropic. The connection between the two strain energies is discussed in Section 3. In Section 4, the class of deformations is restricted to plane strain superposed on uniform extension. Inverse deformation results are derived for transversely isotropic materials and the results of ADKINS [1] for incompressible materials are rederived. Illustrations of the theory are given in Section 5.

2. Basic Equations

We suppose that in a deformation of an elastic body a particle at the point x_i $(i = 1, 2, 3)$ in the initial or reference state is moved to the point y_i with respect to a fixed rectangular cartesian coordinate system. We write

$$y_i = f_i(x_1, x_2, x_3) = f_i(x_r) , \tag{2.1}$$

and we shall use the notations

$$\frac{\partial y_r}{\partial x_s} = y_{r,s} , \quad \frac{\partial x_r}{\partial y_s} = x_{r,s} , \tag{2.2}$$

where Latin indices assume the values 1, 2, 3. We then have

$$y_{i,r}\, x_{r,j} = x_{i,r}\, y_{r,j} = \delta_{ij} , \tag{2.3}$$

where a repeated index implies summation over the values 1, 2, 3, and it follows that

$$\frac{\partial y_{m,n}}{\partial x_{r,s}} = -y_{m,r}\, y_{s,n} , \quad \frac{\partial x_{m,n}}{\partial y_{r,s}} = -x_{m,r}\, x_{s,n} . \tag{2.4}$$

The Jacobian of the deformation is denoted by J and conservation of mass requires

$$J = \det(y_{r,s}) = \frac{\varrho_0}{\varrho} , \tag{2.5}$$

where ϱ_0 and ϱ are the densities in the initial and deformed states respectively.

We shall assume that the body is elastically homogeneous in the initial state. The strain energy W_0 per unit volume of the unstrained state then depends only on the strains e_{ij} defined by

$$e_{ij} = \frac{1}{2}\,(y_{r,i}\, y_{r,j} - \delta_{ij}) \tag{2.6}$$

and we shall write

$$W_0 = W_0(e_{ij}) = F(y_{r,s}) . \tag{2.7}$$

The rectangular cartesian components of stress are given by

$$t_{ij} = \frac{1}{J}\,\frac{\partial W_0}{\partial y_{j,r}}\, y_{i,r} \tag{2.8}$$

and t_{ij} is symmetric since W_0 depends on $y_{r,s}$ only through the strains e_{ij}. With (2.3) the stress-strain relation can also be written

$$t_{ij} = -\frac{1}{J}\,\frac{\partial W_0}{\partial x_{r,i}}\, x_{r,j} . \tag{2.9}$$

The stress-strain relation is modified if geometrical constraints restrict deformations of the body. Here we consider only the constraint of incompressibility which requires

$$J = \det(y_{r,s}) = 1 . \tag{2.10}$$

The stresses are then given by

$$t_{ij} = \frac{\partial W_0}{\partial y_{j,r}} y_{i,r} + p\,\delta_{ij} , \tag{2.11}$$

$$t_{ij} = -\frac{\partial W_0}{\partial x_{r,i}} x_{r,j} + p\,\delta_{ij} , \tag{2.12}$$

where the hydrostatic stress p is a function of position and time.

If we suppose that the deformed body is in equilibrium with no body force, the differential equations of equilibrium are

$$\frac{\partial t_{ij}}{\partial y_i} = 0 . \tag{2.13}$$

These equations can also be written (see [2], for example)

$$\frac{\partial}{\partial x_i}(J\, x_{i,r}\, t_{rj}) = 0 . \tag{2.14}$$

Equation (2.14) can be derived directly from (2.13) by use of the identity

$$\frac{\partial}{\partial x_i}(J\, x_{i,r}) = 0 .$$

3. Inverse Relationships

Substitution of the stress-strain law (2.8) into Equations (2.14) gives the well-known form for the equilibrium equations for an elastic body,

$$\frac{\partial}{\partial x_i}\left(\frac{\partial W_0}{\partial y_{j,i}}\right) = 0 . \tag{3.1}$$

If we multiply this equation by $y_{j,r}$ and interchange j and r we find that

$$\frac{\partial}{\partial x_i}\left(\frac{\partial W_0}{\partial y_{r,i}} y_{r,j}\right) - \frac{\partial W_0}{\partial y_{r,i}} y_{r,ij} = 0 . \tag{3.2}$$

Because of the assumption of elastic homogeneity, W_0 does not depend on x_i explicitly and the second term in (3.2) is therefore $-\partial W_0/\partial x_j$. If we now write

$$W_0 = \frac{\varrho_0}{\varrho} W = J\, W ,$$

so that W is the strain energy per unit volume of the deformed body, then (3.2) becomes

$$\frac{\partial}{\partial x_i}\left(J\,\frac{\partial W}{\partial y_{r,i}} y_{r,j} + W\,\frac{\partial J}{\partial y_{r,i}} y_{r,j} - J\, W\,\delta_{ij}\right) = 0 .$$

However, we have

$$\frac{\partial J}{\partial y_{r,i}} y_{r,j} = A_{ri}\, y_{r,j} = J\,\delta_{ij} ,$$

where A_{mn} is the cofactor of $y_{m,n}$ in $\det(y_{r,s}) = J$. Thus for equilibrium of a homogeneous elastic body we have the alternative form

$$\frac{\partial}{\partial x_i}\left(J\,\frac{\partial W}{\partial y_{r,i}}\,y_{r,j}\right) = 0 \tag{3.3}$$

for the equilibrium equations.

On the other hand substitution of (2.9) into (2.13) leads to

$$\frac{\partial}{\partial y_i}\left(\frac{1}{J}\,\frac{\partial W_0}{\partial x_{r,i}}\,x_{r,j}\right) = 0\,. \tag{3.4}$$

Equations (3.3) and (3.4) are identical in form since $1/J$ is $\det(x_{r,s})$. If we use $W_0 = J\,W$ in (3.4) we get

$$\frac{\partial}{\partial y_i}\left(\frac{\partial W}{\partial x_{r,i}}\,x_{r,j} + \frac{1}{J}\,W\,\frac{\partial J}{\partial x_{r,i}}\,x_{r,j}\right) = 0\,.$$

With the result

$$\frac{\partial J}{\partial x_{r,i}}\,x_{r,j} = -\,J\,\delta_{ij}$$

we arrive at

$$\frac{\partial}{\partial y_i}\left(\frac{\partial W}{\partial x_{r,i}}\right)x_{r,j} + \frac{\partial W}{\partial x_{r,i}}\,x_{r,ij} - \frac{\partial W}{\partial x_j} = 0\,. \tag{3.5}$$

Because W lacks explicit dependence on y_i, the last two terms cancel and with (2.3) we see that (3.5) implies

$$\frac{\partial}{\partial y_i}\left(\frac{\partial W}{\partial x_{j,i}}\right) = 0\,, \tag{3.6}$$

which may be compared with (3.1).

We note that when a body force F_i per unit mass acts, (3.1) and (3.6) are replaced by

$$\frac{\partial}{\partial x_i}\left(\frac{\partial W_0}{\partial y_{j,i}}\right) + \varrho_0\,F_j = 0\,, \qquad \frac{\partial}{\partial y_i}\left(\frac{\partial W}{\partial x_{j,i}}\right) - \varrho\,F_r\,y_{r,j} = 0\,,$$

while the terms $\varrho_0\,F_r\,y_{r,j}$ and $-\,\varrho\,F_j$ are added to the left-hand sides of (3.3) and (3.4), respectively.

For an incompressible material, (2.11) and (2.14) give

$$\frac{\partial}{\partial x_i}\left(\frac{\partial W_0}{\partial y_{j,i}} + p\,x_{i,j}\right) = 0\,, \tag{3.7}$$

while (2.12) and (2.13) give

$$\frac{\partial}{\partial y_i}\left(\frac{\partial W_0}{\partial x_{r,i}}\,x_{r,j} - p\,\delta_{ij}\right) = 0\,. \tag{3.8}$$

From (3.7) we obtain

$$\frac{\partial}{\partial x_i}\left(\frac{\partial W_0}{\partial y_{r,i}}\,y_{r,j} + p\,\delta_{ij}\right) - \frac{\partial W_0}{\partial y_{r,i}}\,y_{r,ij} - p\,x_{i,r}\,y_{r,ij} = 0\,. \tag{3.9}$$

Because $J = 1$ we have

$$x_{i,r}\,y_{r,ij} = A_{ri}\,y_{r,ij} = \frac{\partial J}{\partial x_j} = 0$$

and (3.9) reduces to

$$\frac{\partial}{\partial x_i}\left(\frac{\partial W_0}{\partial y_{r,i}}\,y_{r,j} + (p - W_0)\,\delta_{ij}\right) = 0\,. \tag{3.10}$$

A similar calculation leads from (3.8) to

$$\frac{\partial}{\partial y_i}\left(\frac{\partial W_0}{\partial x_{j,i}} - (p - W_0)\, y_{i,j}\right) = 0\,. \tag{3.11}$$

It may be noted that the term $W_0\,\delta_{ij}$ in (3.10) can be removed by setting $W_0 = J\,W$. Since $J = 1$, W_0 and W are equal in value but their different functional dependence gives

$$\frac{\partial W_0}{\partial y_{r,i}}\, y_{r,j} = \frac{\partial W}{\partial y_{r,i}}\, y_{r,j} + W_0\,\delta_{ij}$$

when J is set equal to unity after differentiation. Similarly the term $W_0\, y_{i,j}$ can be removed from (3.11) but we prefer to use the forms (3.10) and (3.11).

If we interchange x_i and y_i in (3.6) and compare the resulting equation with (3.1) we obtain the following result. If the deformation $y_i = f_i(x_r)$ can be maintained as an equilibrium state without the application of body force for a material with strain energy $W_0 = F(y_{r,s})$, then the inverse deformation defined implicitly by

$$x_i = f_i(y_r) \tag{3.12}$$

is an equilibrium state with no body force for any material which has a strain energy W_0' per unit volume of the unstrained state such that for the deformation (3.12)

$$\frac{\partial W_0'}{\partial y_{j,i}} = \frac{\partial}{\partial y_{j,i}}\{J\,F(x_{r,s})\}\,. \tag{3.13}$$

A multiplicative constant can be included in (3.13).

If the relation (3.13) is satisfied by W_0' for every equilibrium state for the material with strain energy W_0 given by (2.7), so that the inverse of every equilibrium state for W_0 is an equilibrium state for W_0', then we must have

$$W_0' = J\,F(x_{r,s})$$

apart from an additive constant. It follows that W_0' depends on $x_{r,s}$ only through the strains ε_{ij} defined by

$$\varepsilon_{ij} = \frac{1}{2}\,(\delta_{ij} - x_{r,i}\,x_{r,j})\,,$$

since W_0 and J depend only on e_{ij} defined by (2.6). However, the strain energy of an elastic material must be invariant under rigid displacements of the deformed state, which leave e_{ij} unchanged but affect ε_{ij}. Thus W_0' can depend on ε_{ij} alone only when the material is isotropic and this in turn requires the material with strain energy W_0 to be isotropic. In this case W_0 is a symmetric function of the principal extension ratios λ_1, λ_2 and λ_3,

$$W_0 = L(\lambda_1, \lambda_2, \lambda_3)\,,$$

and then W_0' given by

$$W_0' = \lambda_1\,\lambda_2\,\lambda_3\,L\left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}\right) \tag{3.14}$$

will satisfy (3.13). Alternatively W_0 can be considered as a function of three independent strain invariants and we choose these to be

$$\left.\begin{aligned} I = y_{r,s}\,y_{r,s} = \lambda_1^2 + \lambda_2^2 + \lambda_3^2\,, \quad H = x_{r,s}\,x_{r,s} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2} + \frac{1}{\lambda_3^2}\,, \\ J = \det(y_{r,s}) = \lambda_1\,\lambda_2\,\lambda_3\,. \end{aligned}\right\} \tag{3.15}$$

Then if we have

$$W_0 = W_0(I, H, J) , \tag{3.16}$$

the strain energy W_0' given by

$$W_0' = J\, W_0 \left(H, I, \frac{1}{J}\right) \tag{3.17}$$

will satisfy (3.13).

If the two materials are to be the same, so that the inverse of an equilibrium deformation is also a possible equilibrium state for the same material, then we must have $W_0' = W_0$. With (3.16) and (3.17) this requires

$$W_0 (I, H, J) = J\, W_0 \left(H, I, \frac{1}{J}\right),$$

and if we set

$$W_0 = J^{1/2}\, M(I, H, \ln J)$$

we need, for $W_0 = W_0'$,

$$M(I, H, \ln J) = M(H, I, -\ln J) . \tag{3.18}$$

If we now write

$$G_1 = I + H , \quad G_2 = I\, H , \quad G_3 = (I - H) \ln J , \quad G_4 = (\ln J)^2 ,$$

it is apparent that the function

$$M = N(G_1, G_2, G_3, G_4) \tag{3.19}$$

satisfies the invariance property (3.18) for any choice of the function N. Conversely it is easily shown that any function M satisfying (3.18) must be expressible in the form (3.19).

An incompressible material can be treated as a particular case of the preceding or we can compare the equilibrium Equations (3.7) and (3.11) to deduce the following. If the deformation $y_i = f_i(x_r)$ and pressure function $p = k(x_r)$ provide an equilibrium state with no body force for an incompressible material with strain energy $W_0 = F(y_{r,s})$, then the inverse deformation (3.12) is a possible equilibrium state with no body force for an incompressible material which has a strain energy W_0' such that for the deformation (3.12)

$$\frac{\partial W_0'}{\partial y_{j,i}} = \frac{\partial F(x_{r,s})}{\partial y_{j,i}} .$$

The pressure function p' for the inverse deformation is given by

$$p' = -\, k(y_r) + F(x_{r,s})$$

apart from an additive constant. If this result applies for every equilibrium deformation for the material with strain energy W_0, the two materials must be isotropic and then

$$W_0 = W_0(I, H) , \quad W_0' = W_0(H, I) . \tag{3.20}$$

For the two materials to be the same, W_0 must be symmetric in I and H and therefore

$$W_0 = N\,(I + H, I\, H) .$$

4. Plane Strain Superposed on Uniform Extension

Inverse deformation results for anisotropic materials can be obtained if the class of deformations is restricted. In this section we consider plane deformations superposed on a uniform extension normal to the plane so that

$$y_\alpha = f_\alpha(x_\beta; \lambda) , \quad y_3 = \lambda x_3 , \tag{4.1}$$

where λ is a constant and Greek indices assume the values 1, 2. For the deformation (4.1) we have

$$x_{\alpha,\beta} x_{\alpha,\beta} [\det(y_{\gamma,\delta})]^2 = y_{\rho,\mu} y_{\rho,\mu}$$

and the strain invariants (3.15) are given by

$$I = \lambda^2 + \Theta , \quad H = \frac{1}{\lambda^2} + \frac{\Theta}{\Delta^2} , \quad J = \lambda \Delta , \tag{4.2}$$

where we have written

$$\Theta = y_{\alpha,\beta} y_{\alpha,\beta} , \quad \Delta = \det(y_{\alpha,\beta}) .$$

A transversely isotropic material with elastic axis in the x_3-direction has a strain energy of the form (see [3], for example)

$$W_0 = W_0(I, H, J, K_1, K_2) , \tag{4.3}$$

where

$$K_1 = e_{33} , \quad K_2 = e_{3\alpha} e_{3\alpha} ,$$

and we shall assume that W_0 does not depend on x_i explicitly. For the deformation (4.1),

$$K_1 = \frac{1}{2} (\lambda^2 - 1) , \quad K_2 = 0 \tag{4.4}$$

and (4.3) reduces to

$$W_0 = \Sigma(\Theta, \Delta, \lambda) .$$

The stress t_{33} is independent of x_3 while t_{13} and t_{23} are zero and equilibrium implies, from (3.1) and (3.6),

$$\frac{\partial}{\partial x_\alpha} \left(\frac{\partial W_0}{\partial y_{\beta,\alpha}} \right) = 0 , \quad \frac{\partial}{\partial y_\alpha} \left(\frac{\partial W}{\partial x_{\beta,\alpha}} \right) = 0 .$$

We now see that if (4.1) is an equilibrium state for the material with strain energy W_0, then the inverse deformation defined by

$$x_\alpha = f_\alpha(y_\beta; \lambda) , \quad y_3 = \frac{x_3}{\lambda} , \tag{4.5}$$

gives an equilibrium state for any transversely isotropic material with strain energy W' per unit volume of the strained state such that when (4.2) and (4.4) hold W' reduces to

$$W' = \Sigma \left(\frac{\Theta}{\Delta^2} , \frac{1}{\Delta} , \frac{1}{\lambda} \right) .$$

If (4.1) defines an equilibrium state for λ in the range

$$\frac{1}{\Delta} \leqslant \lambda \leqslant \Delta ,$$

we can replace λ by $1/\lambda$ in (4.5) and then

$$x_\alpha = f_\alpha\left(y_\beta; \frac{1}{\lambda}\right), \quad y_3 = \lambda\, x_3\,, \tag{4.6}$$

defines an equilibrium state for the material with strain energy W' per unit volume of the equilibrium state. Inverting the deformation (4.1) in the plane of isotropy only leads to

$$x_\alpha = f_\alpha(y_\beta; \lambda)\,, \quad y_3 = \lambda\, x_3\,, \tag{4.7}$$

and this will be an equilibrium state for a transversely isotropic material with strain energy W'' per unit volume of the strained state such that when (4.2) and (4.4) hold W'' reduces to

$$W'' = \Sigma\left(\frac{\Theta}{\Delta^2}, \frac{1}{\Delta}, \lambda\right).$$

When the material is incompressible, the condition $J = 1$ requires for the deformation (4.1)

$$\Delta = \det(y_{\alpha,\beta}) = \frac{1}{\lambda} \tag{4.8}$$

and (4.2) then gives

$$I = \lambda^2 + \Theta\,, \quad H = \frac{1}{\lambda^2} + \lambda^2\,\Theta\,. \tag{4.9}$$

The strain energy W_0 for a transversely isotropic material with elastic axis in the x_3-direction is now of the form

$$W_0 = W_0(I, H, K_1, K_2) \tag{4.10}$$

and it reduces to

$$W_0 = \Sigma(\Theta, \lambda) \tag{4.11}$$

when (4.1) applies. The equivalent forms (3.7) and (3.11) of the equilibrium equations require p to be independent of x_3 in this case and

$$\frac{\partial}{\partial x_\alpha}\left(\frac{\partial W_0}{\partial y_{\beta,\alpha}} + p\, x_{\alpha,\beta}\right) = 0\,, \quad \frac{\partial}{\partial y_\alpha}\left(\frac{\partial W_0}{\partial x_{\beta,\alpha}} - (p - W_0)\, y_{\alpha,\beta}\right) = 0\,. \tag{4.12}$$

We deduce that if (4.12) hold for the deformation (4.1) with λ between $1/\Lambda$ and Λ, then the inverse deformation (4.5) or (4.6) is an equilibrium state for an incompressible, transversely isotropic material with strain energy W_0' which reduces to

$$W_0' = \Sigma\left(\lambda^2\,\Theta, \frac{1}{\lambda}\right) \tag{4.13}$$

when (4.9) and (4.4) hold. When $\lambda = 1$, (4.11) and (4.13) coincide and the inverse deformation is possible for the same material. This is a special case of a more general result obtained by ADKINS [1] and discussed in the next paragraph. ADKINS also showed that an isotropic material with the Mooney form

$$W_0 = C_1\,(I - 3) + C_2\,(H - 3) \tag{4.14}$$

for the strain energy, where C_1 and C_2 are constants, permits the inverse deformation (4.5) or (4.6) as an equilibrium state when (4.1) is an equilibrium state for the same material. This can be seen here by noting that in this case

$$\Sigma(\Theta, \lambda) = (C_1 + \lambda^2\, C_2)\,\Theta + C_1\,(\lambda^2 - 3) + C_2\left(\frac{1}{\lambda^2} - 3\right) \tag{4.15}$$

and the inverse deformation (4.5) is an equilibrium state for a material such that for deformations of the type (4.1)

$$W_0' = A\,\Sigma\left(\lambda^2\,\Theta, \frac{1}{\lambda}\right) + B\,, \tag{4.16}$$

where A and B are constants. By choosing A and B suitably we can make (4.15) and (4.16) coincide for each value of λ. A similar argument applies when W_0 has the form

$$W_0 = C_1\,(I-3) + C_2\,(H-3) + w(I, H, K_1, K_2)\,,$$

where w reduces to a constant whenever (4.4) holds, so that the inverse (4.5) or (4.6) is an equilibrium state for such a material when (4.1) is an equilibrium state for the same material.

ADKINS [1] has shown that for an incompressible transversely isotropic material with strain energy (4.10), the deformation

$$x_\alpha = f_\alpha(\lambda\, y_\beta; \lambda)\,, \qquad y_3 = \lambda\, x_3 \tag{4.17}$$

defines an equilibrium state whenever (4.1) is an equilibrium state for the same material. The proof of this result depends essentially on the following. For incompressible deformations of the type (4.1) we have

$$\Theta = y_{\alpha,\beta}\,y_{\alpha,\beta} = \frac{x_{\alpha,\beta}\,x_{\alpha,\beta}}{\lambda^2}\,.$$

Thus we may take (4.11) for W_0 in the first equation of (4.12) and take

$$W_0 = \Sigma\left(\frac{x_{\alpha,\beta}\,x_{\alpha,\beta}}{\lambda^2},\ \lambda\right)$$

in the second of (4.12). If we replace $\lambda\, y_\alpha$ by y_α in the latter equation, then, after adjustment of the pressure function p, the first and second equations in (4.17) become identical in form with x_α and y_α interchanged. Further the incompressibility condition (4.8) is transformed into itself by replacing $\lambda\, y_\alpha$ by y_α and then interchanging x_α and y_α. Thus (4.17) will be an equilibrium state if (4.1) is an equilibrium state for the same material.

5. Examples

From the work in Section 3, whenever the deformation (2.1) is an equilibrium state for a homogeneous isotropic material with strain energy W_0 given by (3.16), the inverse (3.12) is an equilibrium state for the material with strain energy W_0' given by (3.17), and the result applies to incompressible materials with W_0 and W_0' given by (3.20). It follows that if (2.1) is an equilibrium state for *all* homogeneous isotropic materials then the inverse is also an equilibrium state for all homogeneous isotropic materials. For compressible materials, the only deformation possible in all homogeneous isotropic materials without the application of body force is a state of homogeneous strain, as shown by ERICKSEN [4]. For incompressible materials, however, a number of equilibrium deformations have been determined by RIVLIN and others using a general form $W_0(I, H)$ for the strain energy. Descriptions of these special deformations can be found in [5, 6, 7, 8], but our result does not increase their number

as the inverse of each deformation is included in the list of known deformations of this type.

A recent paper [9] considered a circular disc of incompressible isotropic material bonded between two parallel rigid end-plates and subjected to tension or compression normal to the end-plates. With the assumption that a particle initially at a point (r, θ, z) in a cylindrical polar coordinate system moves to the point (R, θ, Z), where

$$R = r f(z) , \qquad Z = g(z) ,$$

and $f^2 dg/dz = 1$ for incompressibility, it was found that with the Mooney form (4.14) for the strain energy W_0 the function $f(z)$ can be determined so that the equilibrium equations are satisfied with zero body force.

Solutions were determined explicitly for the particular cases of a neo-Hookean material, for which $C_2 = 0$, and a material for which $C_1 = 0$. When $C_2 = 0$, so that

$$W_0 = C_1 (I - 3) , \tag{5.1}$$

the solution for tension in [9] is

$$R = \lambda r \cosh \alpha z , \qquad Z = \frac{1}{\lambda^2 \alpha} \tanh \alpha z \tag{5.2}$$

in which α and λ are parameters. When $C_1 = 0$, so that the strain energy is W_0' given by

$$W_0' = C_2 (H - 3) , \tag{5.3}$$

the solution in [9] for compression is

$$R = \lambda r \left[1 - \left(\frac{\alpha z}{\lambda}\right)^2\right], \qquad Z = \frac{1}{\lambda \alpha} \tanh^{-1} \left(\frac{\alpha z}{\lambda}\right). \tag{5.4}$$

If the parameters α, λ in (5.4) are adjusted, it can be seen that (5.4) is the deformation inverse to (5.2) and this is in agreement with Section 3 because W_0 and W_0' given by (5.1) and (5.3) satisfy the inverse relation (3.20), apart from a multiplicative constant. Similarly, the solution for the material (5.1) in compression is the inverse of the solution for (5.3) in tension and in fact they are the solutions (5.2) and (5.4) with α replaced by $i\alpha$.

The results derived by ADKINS for incompressible transversely isotropic materials in plane strain were illustrated by examples in [1]. For certain problems, the inverse deformation (4.17) can be the same as the original deformation (4.1) apart from adjustment of a parameter and this has been applied in [10] to reduce the labor in the solution of these problems by the method of successive approximations.

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Zusammenfassung

Es wird gezeigt, dass für elastisch homogenes Material die Gleichgewichtsbedingungen ohne Massenkräfte bezüglich Ausgangs- und Endkoordinaten sowie Deformationsenergie je Raumeinheit im undeformierten und im deformierten Zustand eine Dualität aufweisen. Mit Hilfe dieser Eigenschaft wird gezeigt, dass der Gleichgewichtszustand für ein gegebenes Material eine inverse Verformung definiert, welche für gewisse andere Materialien einen Gleichgewichtszustand darstellt. Die Arbeit verallgemeinert die von ADKINS für transversal isotropes inkompressibles Material in gleichförmigem Zug mit überlagerter ebener Verformung nachgewiesene Reziprozität.

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Thermal Stresses in an Elastic Half-space Associated with an Arbitrarily Distributed Moving Heat Source[1])

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Introduction

When two bodies are in rolling and sliding contact under heavy loads, local high temperature often occurs as a result of excessive frictional heating due to sliding of the boundaries. The characteristics of this temperature field depend not only upon the geometry of the contact but also upon the load, rolling and sliding speed, surface finish and the lubricant used in the contact.

Failure records indicate that life of these heavily-loaded contacts becomes drastically reduced whenever the frictional heating becomes severe in the contact. It is suspected that this reduction in life is related to the additional stresses produced by the large temperature differentials near the contacting surface. In order to make quantitative prediction of the reduction of life of these contacts due to thermal effects, it is

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